QPCODE: **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION EXAMINATION

IV SEMESTER

MG4DSCCFS200-DATA STRUCTURE USING C++

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hrs **Maximum Marks: 50 Marks**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

1. Define the term data structure. [CO:1]
2. List the various operations that can be performed on a data structure. [CO:1]
3. Recall the initial value of the top pointer/index in an array-based stack implementation when the stack is empty. [CO:2]
4. Summarize the concept of stack overflow and underflow conditions. [CO:2]
5. Define a queue data structure and its basic principle. [CO:1]
6. Predict the behavior of a priority queue when multiple elements have the same priority. [CO:2]
7. Identify the primary difference in memory allocation between arrays and linked list. [CO:1]
8. State the purpose of a null pointer at the end of a singly linked list. [CO:1]

9. Illustrate the concept of balance factor in AVL tree. [CO:2]

10. Compare the terms height and depth of a tree. [CO:2]

11. Classify the types of tree traversal methods. [CO1]

12. Recall the main idea behind selection sort and insertion sort. [CO:1]

[1x10 = 10]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **5** marks

13. Differentiate between a linear and a non-linear data structure with a suitable example for each. [CO:1]

14. Illustrate how an array is used to represent a stack in memory. [CO:2]

15. Discuss the various operations performed on a queue data structure. [CO:1]

16. Explain the insertion operation in linked list [CO:1]

17. Compare a complete binary tree and an AVL tree. [CO:2]

18. Identify the differences between binary search with linear search. [CO:1]

[5x4 = 20]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

19. Explain in your own words why an array is considered as a linear data structure with suitable examples. [CO:1]

20. Distinguish between PUSH and POP operations of a stack with suitable algorithms. [CO:2]

21. Explain the working of bubble sort with suitable example. [CO:1]

[10x2 = 20]